various time intervals using historical stock prices and news sentiments. The predicted news sentiments from news articles are joined with stock market data using date.

Methodology

The section covers the implementation details on determining the sentiments conveyed by news articles, and how these sentiments impact stock prices. The methodology is based on the research methodology by (Khedr et al., 2017) with the introduction of fine-grained analysis of input text (Meyer et al., 2017).

The specifications of the machine used for the purpose of the research is shown in Table 3.1. The programming language used for the research is Python and the editor used for developing the algorithm is Jupyter Notebook.

Table 3.1. Machine specifications

Processor	Intel® Core™ i5-8350U CPU @ 1.70GHz
RAM	16.0 GB
System type	X64-based processor
Operating System	Windows 10 Pro

Conceptual Framework

The study proposes the utilization of various fine-grained syntactic Part of Speech (POS) combinations when parsing news articles. The resulting deciphered sentiments are used together with historical opening, highest, and lowest stock market prices for each day in determining the adjusted closing price of stock indices. The stock indices that are being looked into are the top 5 companies that account for 17.5% of S&P 100, which are Apple, Microsoft, Google, Amazon, and Facebook (Garcia-Medina et al., 2017). The proposed model takes two set of inputs; the first set comprises the sentiments obtained from sentiment analysis model, and the second set contains the stock market historical prices. The stock prediction model looks into predicting the adjusted closing price in various time intervals using historical stock prices and news sentiments. The predicted news sentiments from news articles are joined with stock market data using date.

The algorithm is comprised of multiple steps, namely data preprocessing, Part-of-Speech (POS) tagging, conversion into vector representation, application of machine learning classifiers, and joining the output sentiments with stock market data using date and feeding the joined dataset into K-NN algorithm. The following sections expound on the details of the model. A summary of the model is outlined in Figure 3.1.

News Sentiment Analysis

Data gathering

The news articles are obtained through New York Times (NYT) API. The study conducted by Garcia-Medina et al. used dataset from NYT and the results indicate a significant correlation between news and stocks. In addition, as reported by Forbes, the revenue of NYT has increased from 2016 to 2018 by about 12%, and one of the reasons is the boost in subscriptions. This depicts that increasing number of people are relying on the news feeds of NYT which makes it a good source of data for this research. The number of articles obtained per day is limited to three (3) as is also done in (Khedr et al., 2017).

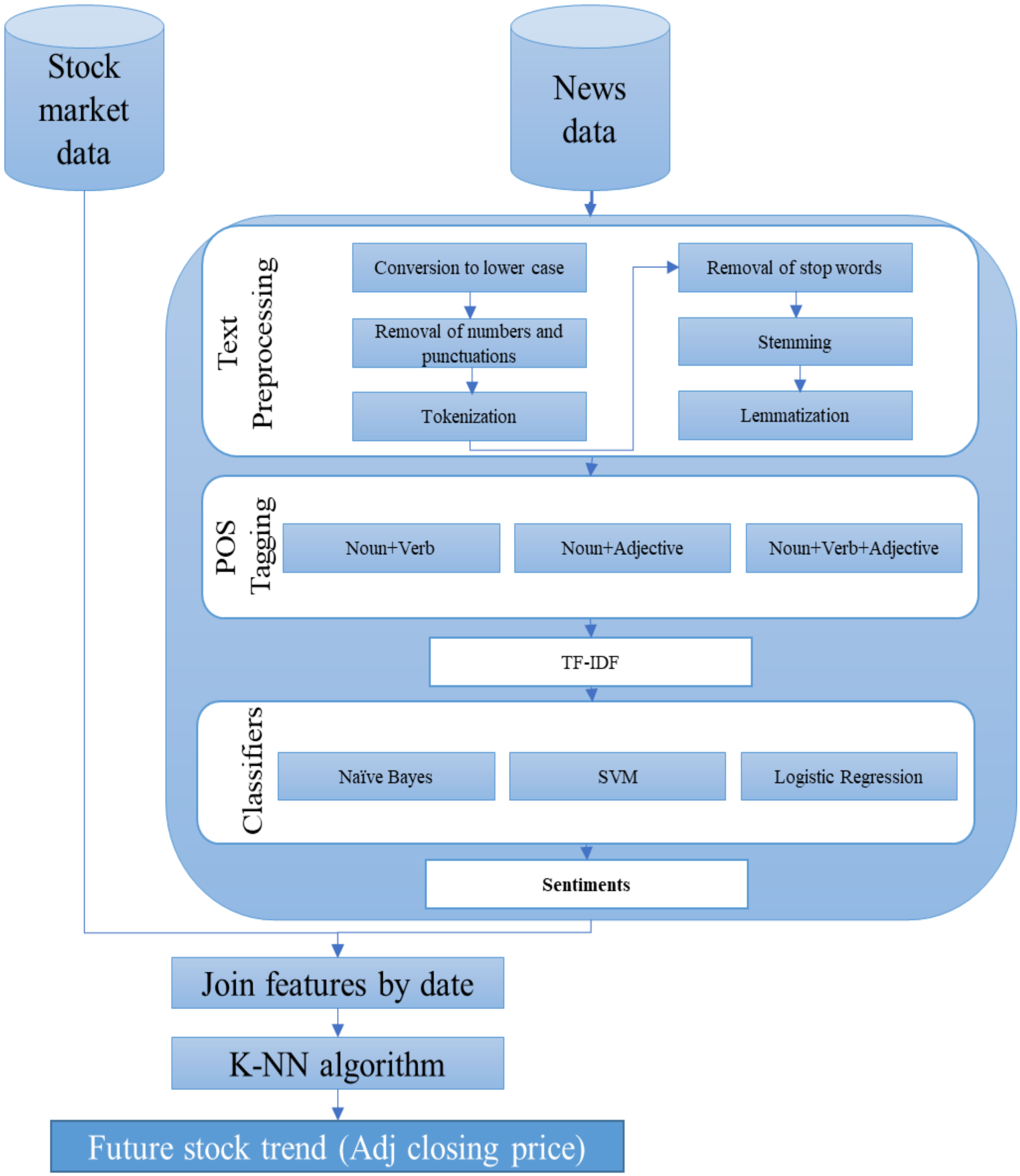

Figure 3.1. Conceptual framework

The archive of news ranges from the 1st of January, 2013 until 31st of December, 2017. Figure 3.2 depicts a summary of the data gathering process. Firstly, the researcher signed up in NYT Developer website in order to get an API key. A python program is developed to gather data from NYT website. The program sends HTTP request to NYT containing article date and API key parameters. The response received is in JSON form. The program parses the response and goes through each of the articles in the parsed content. The program also makes sure to collect articles only from 1st of January, 2013 until 31st of December, 2017.

Gathered data information extraction

Every article is being checked if it falls under Business day, Business, Entrepreneurs, or Financial category. If the category falls under any of the aforementioned sections, then the article headline is being looked into to see if there is a named entity categorized under Organization ("ORG"). Named Entity Recognition (NER) classifies entities into pre-defined categories which include persons, locations, organizations, expressions of times, etc. The purpose of collecting only those articles having headlines that contain organization entity is to retrieve articles that convey news about companies. Only three (3) news articles are collected per day, which means that the first three (3) articles gathered from the parsed JSON response which satisfy the article category and named entity category are added to the list. After gathering, the articles are then annotated by the stock investor into positive and negative. Table 3.2 depicts a small sample of gathered news dataset with the annotations.

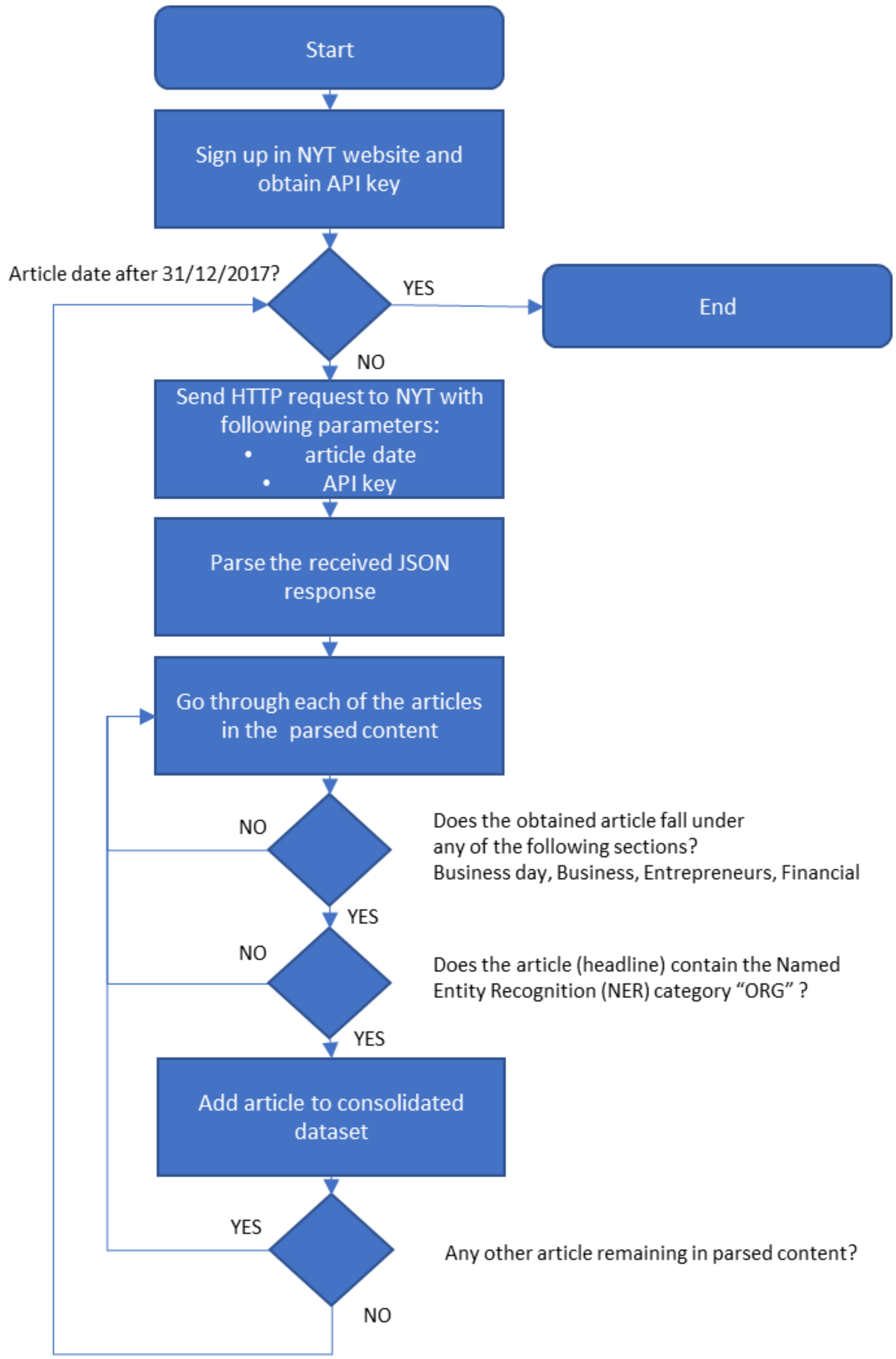

Figure 3.2. Data gathering process flow

Table 3.2. Sample gathered news dataset consisting of date, stock investor annotations, and news *(only first sentence of the articles is shown below)*

Date	Annotation	News 1	News 2	News 3
01/01/2013	1	The 2013 Chevrolet Cruze comes with a new air bag technology: a frontal air bag for the driver that is designed with a different type of venting.	DETROIT Ford Motor, the nation No. 2 automaker, said on Tuesday that its fourth-quarter earnings rose 90 percent, to $3 billion, mostly because of favorable tax benefits related to investments in its European operations.	The pink Toyota Crown sedan that took center stage at a holiday event here last week was meant to shock, and it did.
02/01/2013	1	A 2012 BMW luxury car from the 5 Series on the production line of the German car manufacturers plant in the Bavarian city of Dingolfing in March.	SAN FRANCISCO For 19 months, Google pressed its case with antitrust regulators investigating the company.	Time Warner reported on Wednesday that its profits in the fourth quarter shrank by 12 percent, citing higher programming costs at its Home Box Office and Turner units and reinvestments in the business that cut into the bottom line.
03/01/2013	-1	Former Secretary of Defense Robert M. Gates congratulate Senator Hagel on his nomination to be secretary of defense.	The $3.49 billion endowment fund at Dartmouth, an Ivy League college in Hanover, N.H., returned 5.8 percent for the 12 months that ended in June.	AUTOMAKERS are racing to ramp up their digital offerings, linking in-dash systems to smartphones and services like traffic monitors faster than you can say Pandora.
04/01/2013	1	Among the panoply of gizmos and gadgets taking bows this week at the Consumer Electronics Show in Las	Fiat Chrysler is recalling some 25,500 of its 2014-15 Fiat 500L models	How dead is the Fox-Time Warner deal? So dead that Rupert Murdoch hopped

		Vegas, relatively few are intended to keep you safe in an automobile even in an automobile that driving itself.	because a knee air bag designed to protect the driver legs in a frontal impact may malfunction, according to a report from the automaker posted on the website of the National Highway Traffic Safety Administration.	on his company quarterly earnings call on Wednesday to drive home the point that his giant media company, 21st Century Fox, really is walking away.
05/01/2013	1	Herbalife, a maker of nutritional supplements and protein bars, recruits others to do its selling.	The Gannett Company reported a major rise in third-quarter profit on Monday, largely because of an infusion of political advertising at its television stations, which nearly doubled in number after the absorption of the Belo Corporation stations bought last year.	Ford is recalling about 745,000 sedans and sport utility vehicles in the United States for a problem with an electronic module that could prevent the air bags from deploying in a crash, the automaker said Friday.

Data preprocessing

In order to make the data suitable for further processing, there is a need to clean it first. The data preprocessing comprises of converting the text into lowercase, removing of numbers and punctuation marks, tokenization, removal of stop words (e.g. ours, you're, himself, her, against, etc.), stemming or reducing a word into its word stem by cutting off the prefix or suffix of the inflected word, and lemmatization or converting the word into its base form through morphological analysis. The python libraries primarily used for data preprocessing are **spacy** and **nltk** (Mo et al., 2016). Table 3.3 depicts the preprocessed gathered news dataset.

Table 3.3. Preprocessed gathered news dataset
(only first sentence of the articles is shown below)

Date	Annotation	News 1	News 2	News 3
01/01/2013	1	chevrolet cruze come new air bag technology frontal air bag driver designed different type venting	detroit ford motor nation automaker said Tuesday fourth quarter earnings rose percent billion mostly favorable tax benefit related investment European operation	pink toyota crown sedan took center stage holiday event last week meant shock
02/01/2013	1	bmw luxury car series production line german car manufacturer plant bavarian city dingolfing march	san francisco month google pressed case antitrust regulator investigating company	time warner reported wednesday profit fourth quarter shrank percent citing higher programming cost home box office turner unit reinvestments business cut bottom line
03/01/2013	-1	former secretary defense robert gate congratulate senator hagel nomination secretary defense	billion endowment fund dartmouth ivy league college hanover returned percent month ended june	automaker racing ramp digital offering linking dash system smartphones service like traffic monitor faster say pandora
04/01/2013	1	among panoply gizmo gadget taking bow week consumer electronics show la vega relatively intended keep safe automobile even automobile driving	fiat chrysler recalling fiat l model knee air bag designed protect driver leg frontal impact may malfunction according report automaker posted website national highway traffic safety administration	dead fox time warner deal dead Rupert murdoch hopped company quarterly earnings call wednesday drive home point giant medium company century fox really walking away
05/01/2013	1	herbalife maker nutritional supplement protein bar recruit others selling	gannett company reported major rise third quarter profit Monday largely infusion political advertising television station nearly doubled number absorption belo corporation station bought last year	ford recalling sedan sport utility vehicle united state problem electronic module could prevent air bag deploying crash automaker said friday

Conversion into lowercase and tokenization

Each of the words in the text are converted into lower case and are then tokenized. Tokenization is the process of dividing or splitting a given text into small units or tokens. The tokens can be in the form of words, numbers, punctuation marks, etc. Each of the news entries are tokenized into a series words.

Removal of stop words, numbers and punctuation marks

After tokenization, the stop words are removed (e.g. in, after, on, you, etc.). These are removed since they occur in abundance in text, thereby providing little to no unique information or clue about the given text. The numbers (0-9) are also removed since these are not needed. Sample numbers are dates, percentages, amounts, etc. In addition, punctuation marks are removed as well.

Stemming

In stemming, the tokenized words are reduced to their word roots or stems. It works by either removing the prefix or suffix of the word. However, since stemming is based on heuristics, it, alone, is not enough for getting the root form of words.

Lemmatization

Lemmatization is the process of getting the morphological form of the word. It aims to remove the inflectional endings of words and returns the based or dictionary form of these words. After applying stemming on the words, lemmatization is performed as well.

POS tagging

Each of the words in the preprocessed text are then marked to their corresponding part-of-speech. The tagging of words is based on the process employed by (Meyer et al., 2017) in their study. Different combinations of noun, adjective, verb, and company name (a custom named entity that is a proper noun and refers to the company name) are used. The purpose of this is to determine which combination is optimum or contains the most amount of information that could be used for text analysis. In addition, this can somehow diminish the text to be processed in the next steps of the algorithm, thereby helping to reduce the overall processing time. For the purpose of this study, the following combinations will be analyzed: Noun (Company Name) + Verb, Noun (Company Name) + Adjective, and Noun (Company Name) + Adjective + Verb. The python library used for the purpose is nltk (Meyer et al., 2017). Table 3.4, Table 3.5, and Table 3.6 depicts the resulting news dataset when Noun + Verb, Noun + Adjective, and Noun + Verb + Adjective combinations are used, respectively.

Table 3.4. Preprocessed gathered news dataset with Noun + Verb combination
(only first sentence of the articles is shown below)

Date	Annotation	News 1	News 2	News 3
01/01/2013	1	chevrolet air bag technology air bag driver type venting	ford motor nation automa ker said quarter earnings rose tax benefit investment operation	pink sedan took center stage event week meant shock
02/01/2013	1	luxury car series production line car manufacturer plant city dingolfing march	month google pressed case regulator investigating company	time warner reported profit quarter shrank citing programming cost home box office turner unit reinvestments business cut line
03/01/2013	-1	secretary defenserobert g ate congratulate senator nomination secretary defense	fund dartmouth league college hanover returned month ended june	automaker racing offering linking dash system service traffic monitor pandora
04/01/2013	1	gadget taking week consumer electronics automobile driving	chrysler recalling l model knee air bag driver leg impact according report automaker posted highway traffic safety	fox time deal Rupert murdoch hopped company earnings call home point giant medium company

			administration	century fox walking
05/01/2013	1	herbalife maker supplement protein bar recruit others selling	gannett company reported rise quarter profit monday advertising television station doubled number absorption corporation station bought year	ford recalling sport utility vehicle state problem module air deploying crash automaker said friday

Table 3.5. Preprocessed gathered news dataset with Noun + Adjective combination
(only first sentence of the articles is shown below)

Date	Annotation	News 1	News 2	News 3
01/01/2013	1	chevrolet come new air bag technology frontal air bag driver different type venting	detroit ford motor nation automaker tuesday fourth quarter earnings percent favorable tax benefit investment european operation	pink toyota crown sedan center stage holiday event last week shock
02/01/2013	1	bmw luxury car series production line german car manufacturer plant bavarian city march	san francisco month google case antitrust regulator company	time warner wednesday profit fourth quarter percent higher cost home box office turner unit reinvestments business bottom line
03/01/2013	-1	former secretary defense robert gate congratulate senator hagel nomination secretary defense	endowment fund dartmouth ivy league college hanover percent month june	automaker ramp digital offering dash system smartphones service traffic monitor pandora
04/01/2013	1	panoply gizmo gadget bow week consumer electronics intended safe automobile	fiat chrysler fiat l model knee air bag driver leg frontal impact report automaker website national highway traffic safety administration	dead fox time warner deal dead rupert murdoch company quarterly earnings call drive home point giant medium company century fox
05/01/2013	1	herbalife maker nutritional supplement protein bar recruit others	gannett company major rise third quarter profit monday infusion political advertising television station number absorption belo corpo ration station last year	ford sedan sport utility vehicle united state problem electronic module air crash automaker friday

Table 3.6. Preprocessed gathered news dataset with Noun + Verb + Adjective combination *(only first sentence of the articles is shown below)*

Date	Annotation	News 1	News 2	News 3
01/01/2013	1	chevrolet come new air bag technology frontal air bag driver different type venting	detroit ford motor nation automaker said tuesday fourth quarter earnings rose percent favorable tax benefit investment european operation	pink toyota crown sedan took center stage holiday event last week meant shock
02/01/2013	1	bmw luxury car series production line german car manufacturer plant bavarian city dingolfing march	san francisco month google pressed case antitrust regulator investigating company	time warner reported wednesday profit fourth quarter shrank percent citing higher programming cost home box office turner unit reinvestments business cut bottom line
03/01/2013	-1	former secretary defense robert gate congratulate senator hagel nomination secretary defense	endowment fund dartmouth ivy league college hanover returned percent month ended june	automaker racing ramp digital offering linking dash system smartphones service traffic monitor pandora
04/01/2013	1	panoply gizmo gadget taking bow week consumer electronics intended safe automobile driving	fiat chrysler recalling fiat l model knee air bag driver leg frontal impact according report automaker posted website national highway traffic safety administration	dead fox time warner deal dead rupert murdoch hopped company quarterly earnings call drive home point giant medium company century fox walking
05/01/2013	1	herbalife maker nutritional supplement protein bar recruit others selling	gannett company reported major rise third quarter profit monday infusion political advertising television station doubled number absorption belo corporation station bought last year	ford recalling sedan sport utility vehicle united state problem electronic module air deploying crash automaker said friday

Vector Representation

In order to transform text into a vector, Term Frequency – Inverse Document Frequency (TF-IDF) is used. It depicts the important attributes of the input text. The vector representation gives high values for a term if the term occurs frequently in in the specific document, yet very seldom elsewhere. Meanwhile, if the same term occurs in all documents, the TF-IDF would be 0 (Kalyani et al., 2016). Equation (1) depicts the calculation of TF-IDF, where $tf_{i,j}$ is the number of occurrences of i in j, df_i is the number of documents containing i, and N is the total number of documents.

$$TF - IDF = tf_{i,j} \ x \ log\left(\frac{N}{df_i}\right) \tag{1}$$

Sentiment analysis model

For classification of the inputs, the machine learning classifiers to be used are Logistic regression, Support-Vector Machine (SVM), and Naïve Bayes classifier. As Khedr et al. indicated in their study, these algorithms perform well in text classification.

Naïve Bayes is a probabilistic machine learning model that fits tasks in relation to classification. Equation (2) shows the Bayes theorem where A is hypothesis and B is evidence. Using the theorem, the probability of A happening can be determined given that B has occurred. The assumption of the theorem is that the features are independent such that the presence of one predictor does not affect the other; hence, the term naïve. The input parameter values used for Naïve Bayes are as follows: alpha = 0.001 and binarize = 0.001.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{2}$$

SVM, on the other hand, aims to find a hyperplane that has the maximum margin in an n-dimensional space in order to distinctly classify the data points. Hyperplanes are the decision boundaries that helps in the classification of data points. Data points that fall on whichever side of the hyperplane can be attributed to different categories. The hyperplane dimension is dependent on the number of features. For instance, if the number of input features is 2, then the hyperplane would just be a line. Meanwhile, the hyperplane would be a two-dimensional plane if the number of input features is 3. Figure 3.3 depicts the hyperplanes used for input features equal to 2 and 3. The input parameter values used for SVM are as follows: C = 1, loss = 'squared_hinge', and penalty = 'l2'. The regularization parameter C is chosen to be 1 due to exhaustive grid search (discussed in succeeding section). On the other hand, loss and penalty values are the default ones.

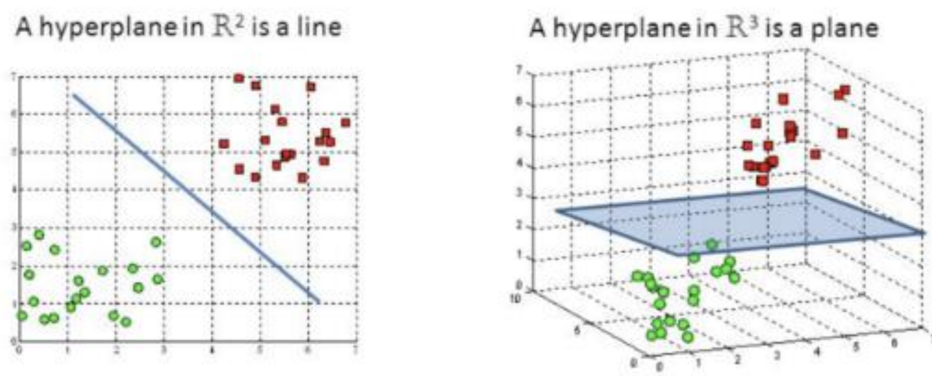

Figure 3.3. SVM hyperplane when input features are 2 (left) and input features are 3 (right) [50]

Logistic regression produces a logistic curve which is created using natural logarithm of the 'odds' of target variable. Equation (3) depicts the logistic regression formula, where P is the probability of a 1, e is the base of natural logarithm, and a and b are the parameters of the model [15]. The input parameter values of logistic regression are as follows: penalty = 'l2' and dual = True.

$$P = \frac{e^{a+bX}}{1 + e^{a+bX}} \tag{3}$$

Training and validation

For training, news data from 1st of January, 2013 until 31st of December, 2016 is used for training. Cross validation is used to split the input training dataset. There is a need to validate the model so as to make sure that all possible patterns in the input dataset are unveiled during training. Cross validation is used to indicate how well the model will generalize to an independent or an unseen data. Stratified K-Folds cross-validator is used for this purpose.

Stratified K-fold cross-validation

Stratified K-Folds cross-validator is used for validation purposes. In this process, the data is divided into k subsets. At each of these k times, one of the k subset of data is used as the validation set and the other k-1 subsets are combined to form a training set. Stratified K-Fold cross validation is used to ensure that each fold contains about the same proportion of samples of each target class or the mean response value is roughly equal in all folds. Since the news dataset is comprised of imbalanced sentiment annotations, Stratified cross-validator is a better option than mere K-fold. The number of folds used is 5 as it is the default.

Grid search

In order to tune input parameters of the estimator, Grid Search has been used. It is an exhaustive search over various input parameters so as to optimize hyperparameters of the model which should result to the best or most accurate predictions. It exhaustively generates candidates from a grid of parameter values that are included in param_grid parameter of the GridSearchCV python function. In the code, the regularization parameter C is exhaustively being searched for. The parameter is used to generalize the classifier to unseen data. It controls the tradeoff between achieving a low training error and a low testing error. The values specified in C parameter are 1, 10, 100, and 1000 (param_grid = {'C' : [1, 10, 100, 1000]}). After running grid search, the optimum value obtained of C is 1. As for scoring, ROC AUC is used.

Testing

For testing, the news data from 1st of January, 2017 until 31st of December, 2017 is used, which is used as input for stock price prediction in the latter part. The metrics to be used for evaluation of results is Area Under the Receiver Operating Characteristic Curve (AUC), Accuracy, and F1-score.

AUC is the probability that a positive instance would be ranked higher than a negative instance by the given model. Equation (4) depicts how AUC is computed, where False Positive Rate (FPR) is FP/(FP+TN) , True Positive Rate (TPR) is TP/(TP+FN) , X_{neg} is the negative instance score, and X_{pos} is the positive instance score [15]. In our analysis, we're looking more into AUC as it is statistically proven to be more consistent and discriminating than accuracy [47,48].

$$\begin{aligned} AUC &= \int_{\infty}^{-\infty} TPR(T)FPR'(T)dT \\ &= P\left(X_{pos} > X_{neg}\right) \end{aligned} \tag{4}$$

Accuracy is a metric that determines if the given model predicts correctly the labels, and is shown in (5), where TP is True Positive, TN is True Negative, FP is False Positive, and FN is False Negative.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

F1-score is the mean of recall and precision, as depicted in (6), where precision is TP/(TP + FP) and recall is TP/(TP + FN).

$$F_1 = \frac{precision\ x\ recall}{(precision + recall)} \tag{6}$$

Stock price analysis

Data gathering

As for the stocks, the stock market data is gathered using Yahoo! Finance python library, namely yfinance. The stock indices that are being looked into are the top 5 companies that account for 17.5% of S&P 100, which are Microsoft Corporation (MSFT), Facebook, Inc. (FB), Amazon.com, Inc. (AMZN), Alphabet Inc. (GOOG), and Apple Inc. (AAPL) [11] starting 1st of January, 2013 until 31st of December, 2017. The opening price, adjusted closing price, highest price, and lowest price of each day is obtained starting 1st of January, 2013 until 31st of December, 2017 [49]. Table 3.7 depicts a small sample of gathered stock dataset.

Table 3.7. Sample gathered stock dataset consisting of date, open, high, low, and adjusted closing price.

Date	Open	High	Low	Adjacent Closing Price
02/01/2013	256.08	258.1	253.26	257.31
03/01/2013	257.27	260.88	256.37	258.48
04/01/2013	257.58	259.8	256.65	259.15
07/01/2013	262.97	269.73	262.67	268.46
08/01/2013	267.07	268.98	263.57	266.38

Data preprocessing

Based on the obtained stock price features, the percent change between highest and lowest stock price for each day is calculated. Also, the input parameters, which are the opening, highest, and lowest stock prices, are scaled to standardize the dataset.

Prediction model

The proposed model looks into predicting the adjusted closing price in various time intervals using historical stock prices and news sentiments, as depicted in Figure 3.4. The predicted news sentiments from news articles are joined with stock market data using date, as has been shown in Fig. 3.1.

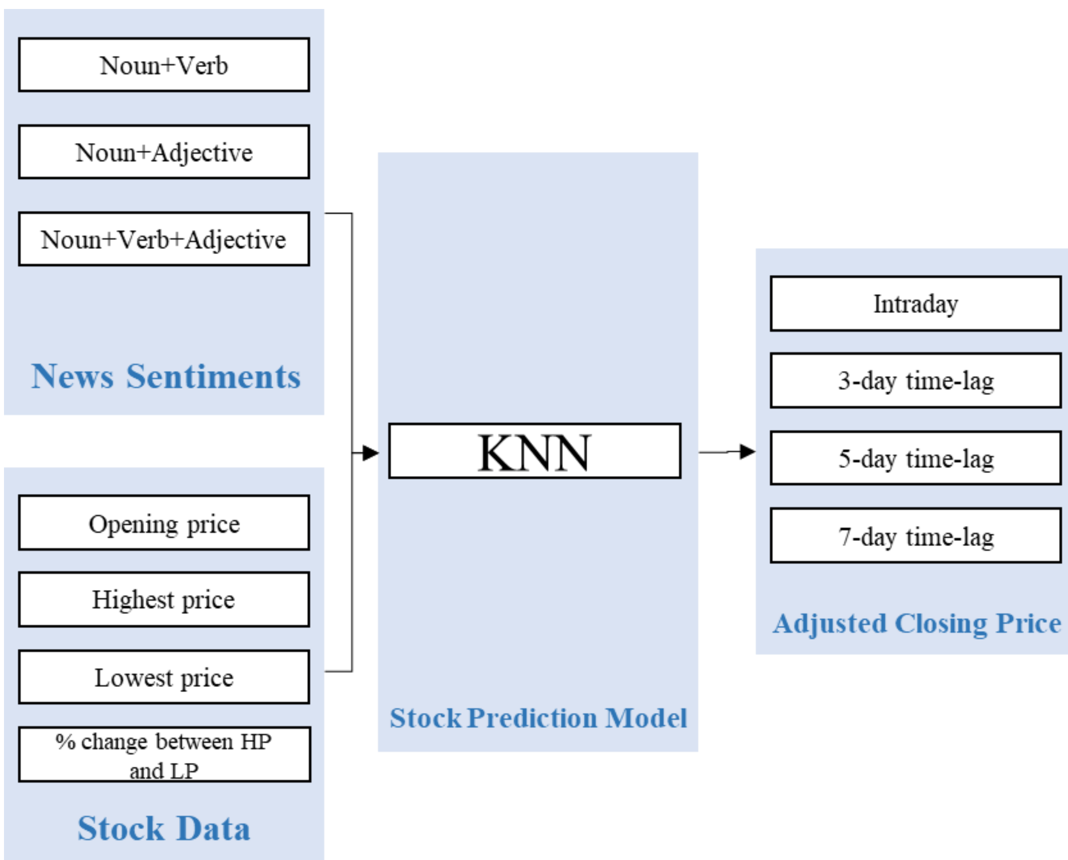

Figure 3.4. Set of inputs and expected outputs of the stock prediction model

In order to predict the stocks, K-Nearest Neighbors (KNN) is used. The algorithm works by calculating the distance between test data and each of the rows in training data using a distance formula, typically the Euclidean distance as shown in Equation (7). Based from the sorted array of training data, the top K rows are chosen and a class is assigned to the test point based on the common class of the rows.

$$d(p,q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \tag{7}$$

Prediction of adjusted closing price in various time-lags

The stocks are predicted for various time-lags. Intraday, 3-day time-lag, 5-day time-lag, and 7-day time-lag stock preidctions are made. Table 3.8 illustrates the inputs to be used for the stock prediction model, which include the sentiments and stock open, high, and low price. The output prediction comprises of stock adjacent closing price for intraday, 3-day, 5-day, and 7-day time spans. The table depicts how shifting in the training dataset is done for various time intervals. For instance, for date 10/01/2013, the sentiment and stock open, high, and low prices for the same date are used and the adjusted closing price for that day is used as well for intraday stock prediction model. For training the model that looks into predicting stocks after 3 days (inclusive), the adjusted closing price is shifted up by 3. Hence, the adjacent closing price of 265.34 (highlighted in green) for date 10/01/2013 has been shifted up by 3 which means the input stocks and sentiment for date 08/01/2013 are matched to the adjacent closing price for 10/01/2013 since ideally, inputs for date 08/01/2013 should be able to predict adjacent closing price for 10/01/2013 for 3-day time-lag prediction. Similarly, for 5-day time-lag prediction, 265.34 from date 10/01/2013 has been moved up to date 04/01/2013 so that inputs for the latter date would be matched to the adjusted closing